

Exploring the Eco-Digital: Performative Sensing with Plants and Data

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ABSTRACT

Artistic practices influenced by philosophical understandings that extend ‘beyond the human’, such as posthuman and post-Anthropocene approaches, partially connect on a qualitative level with ancient understandings of an expanded nature that engulfs both human and nonhuman kin. Preserved in Indigenous epistemes, such as mātauranga Māori (Māori knowledge), this research brings selected aspects of this complex knowledge system into contact with artistic practice generated with networked technologies, to manifest speculative methods of data visualization and signal flow. While there are many different approaches of the eco-digital currently at play, this research looks at practices engaging plants as co-composers. Articulating various situational assemblages that entangle humans, plants, and data, this paper explores theoretical positions that cross-pollinate natural ecology and digital technology. We do this through the prism of two iterative works by the authors: *Contact/Sense* (2019) and *Nga manawataki o te koiora: Biorhythms* (2022).

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1 INTRODUCTION: FRAMING THE ECO-DIGITAL

Embroiled in the eco-digital is the re-assembly of two forces separated by Enlightenment thought: nature and culture. Re-figuring this divide, Donna Haraway pioneered *nature-cultures* as a connection strategy that might re-entangle the two spheres as relational, and further suggests new modes of human-animal relations via the notion of ‘companion species’ [Haraway, 2003]. More recently, Karen Barad’s *agential realism* articulates the situated agency of the nonhuman through perpetually evolving intra-actions that connect subatomic processes to visible materiality [Barad, 2008]. From a post-Anthropocentric perspective, Rosi Braidotti articulates the Posthuman as an entangled process ontology that re-connects Enlightenment dualisms [Braidotti, 2006].

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Important touchstones for this research are artists working intensively with plants as co-composers, such as Christa Sommerer and Laurent Mignonneau’s seminal installation *Interactive Plant Growing*, or Gregory Lasserre and Anais met den Ancxt’s *Akousmaflores*, since both engage tactility to investigate the artistic potential for plants and humans to co-compose together [Sommerer and Mignonneau, 1992, Lasserre and met den Ancxt, 2007]. However, the most continuously resonant examples, are Miya Masaoka’s plant-human interfaces, designed in collaboration with Thomas Zimmerman. In the iterative performance series, such as *Pieces for Plants*, Masaoka networked a specially prepared philodendron by attaching electrodes to its leaves, to react in concert with electrical fields around her body. Working with Zimmerman, she established a body area network (BAN) to sense proximate physiological electrical data, which was then activated through live performance. Using the BAN as an unstable formation, Masaoka modulated with the emergent energy fields coming from the plants. Masaoka has shared her perspective on plants as responsive entities, stating “working with plants in my studio, I was astonished by their ability to respond consistently to my walking in and out of the room ... I [became] increasingly aware of the sensitivity of the plant, and gained greater empathy and awareness of their behavior, needs and responses” [Masaoka, 2018]. Clearly, Masaoka considers the responses and behavior from plants as an element that affectively co-composes the sound piece, citing their ability to emit different signals according to their environment.

As we enter the post-Anthropocene, attention turns to practiced approaches that gesture beyond the human, engaging empathic approaches that situate humans within a wider ecological web. Drawing on co-author Rewa Wright’s cultural heritage as Māori from Aotearoa, infused with select Western philosophical thought, such as posthuman thinking and agential realism, this paper proposes an emergent approach that might re-assemble human relations with plants, through data visualization, sonification, and body-plant sensing. The eco-digital in our practice, articulates a series of techniques and methods that can be further explored through transdisciplinary connections across art, culture and science, traversing both Western epistemology and Indigenous thought. Indigenous thought, framed through this lens, comes specifically from the directions suggested by Rewa’s cultural heritage as a First Nations Māori artist, from Ngai Tawake, Te Kaimaroke, and Te Uri o Hau hapu of Aotearoa/New Zealand. Her ancestry recognizes human and plant relations going back thousands of years, and her personal pepeha (genealogy) links together the rivers, waters, and lands of her traditional rohe (regions), with her human ancestors in a meshwork of relational connections that speak to the entanglements of human and nonhuman kin. Dynamic connections embedded in pepeha, become a lens that allows for patterns of diffraction that move with and through past, present and future. For example, in

Rewa's pepeha, on the Te Uri o Hau line, she traces ancestry to the maunga (mountain) called Tokatoka, the awa (river) called Wairoa, and the moana (sea) called Kaipara. These connections are at the base of her identity, and this is the same for all Māori, since our genealogy is tied to the whenua (land) from birth to death. Transducing this physical connection with the land into computational space is only ever partial, however hints at the ways Indigenous knowledge might be embodied as data and algorithms, nurturing a framework that advances Decolonial thought and gestures toward an Indigenous future.

One way to situate human-nonhuman connections, is to trace methodologies that extend 'beyond human'. Describing a complex web of potential connections or modes of becoming with 'others', the term 'beyond human' emerges notably in the writing of Eduardo Kohn, through his post-anthropological study of the Runa people in Ecuadorian Amazon [Kohn, 2013]. Applied to co-compositional networks of artistic entanglement, 'beyond human' creativity somewhat describes the intuitive desire, but also the technical and creative methods for working with plants and algorithms that Rewa Wright has developed through artistic experiments since 2016. Co-composing beyond the human, is both post-Anthropocentric and post-Anthropological, in Eduardo Kohn's sense of the term. Through research that bridges plant signaling with augmented and extended reality, machine learning, and interaction design for virtual environments, both *Contact/Sense* (2019) and *Nga manawataki o te koiora: Biorhythms* (2022), explore entanglements between human creativity, computational networks and living ecologies [Wright, 2019, Wright and Howden, 2022]. Such an approach aims to destabilize principles of human-centered design, and question the intentionality of the human creator. A 'beyond human' approach to thinking and making allows for an open space of generative production, where unplanned phenomena can unfold, shift, and disrupt each new iteration. Following this thread, Karen Barad's *agential realism* pays attention to the actual movements between types of matter in the world (which she terms 'intra-actions'), the structures or apparatuses that channel matter in a particular configuration, and an understanding of diffraction from Donna Haraway [Haraway, 1997]. A methodological inflection generated by a diffraction of physics and philosophy, agential realism affords an interesting weaving together of the empirical with the only partially knowable.

2 AN INDIGENOUS PERSPECTIVE ON QUANTA

Through imaging technologies such as electron microscopes, quantum physics has been slowly proving to itself that all particles in the universe are deeply entangled. Now, the curious situation exists where Western science has 'validated' what mātauranga Māori (Māori knowledge) has known and preserved for thousands of years. In the Māori world view, people and nature are inseparable and share genealogy. This resonates with current perspectives in evolutionary plant biology, where Mancuso and Viola and others argue that plants are more than 'passive machines' and indeed possess situated modes of 'intelligence' [Mancuso and Viola, 2015]. In the cultural context of mātauranga Māori, such concepts as plant intelligence are well known, since for thousands of years we had

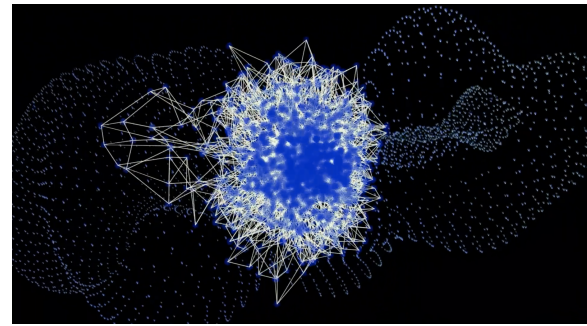


Figure 1: Screen view: *Nga Manawataki o te Koiora* (excerpt image), 2022 version.

preserved intuitive knowledge of deep entanglements between human and nonhuman kin. Embodied in energetic phenomena such as *wairua* (life force shared between all living things) and *mauri* (transitory and shifting states of being), genealogical connections between plants, humans and other animals exist in a contiguous ecological web. Interestingly for this research, the concept articulated as *mauri* is also expressed to a degree in quantum physics. Taina Pohatu has indexed the different states through which *mauri* operates, from sleeping to dynamic, hinting at the vibrancy of *mauri* as it moves through matter [Pohatu, 2011]. Through a diffractive lens, we might say that the different states of *mauri* resonate with scientific evidence within quantum physics that nano level particles indeed bind together every entity in the universe. While 'beyond human' approaches grounded in philosophical re-assembly of Western epistemology are radical in themselves and provide much of value to artistic practices that explore companionship with nonhuman kin (in this research, plants and algorithms), still, they mostly exclude serious consideration of unseen and unknowable cosmological dimensions.

Returning to art, a major challenge within this research is to assemble a hybrid methodology that might selectively weave together anti-colonial modes of knowledge (from Western epistemes), with ancient ways of knowing that privilege sensing, intuition, and spirituality (from Indigenous thinking), while avoiding the more common instrumentalist layering of tradition with digital techniques. Backgrounded by these concepts, we activate a co-collaborative and iterative working method and a series of intra-active techniques with living plants, and custom-built software.

3 CONTACT/SENSE: PERFORMATIVE INTERFACING WITH PLANTS, DATA AND SIGNALS

Approaches that facilitate understandings of the unpredictable and iterative processes behind plant generated data, also act as point of resonance for iterative performance based artistic practices engaging the eco-digital. 'Performative interfacing' is a technique developed by Rewa Wright since 2016, that generates by entanglements between phenomena that emerge by way of gestural apparatuses and carefully configured networks of materials – such as the signals generated by code and plants – that ripple across analogue and digital, as well as physical and screen-based sites [Wright, 2019].

In *Contact/Sense*, performative interfacing is developed by applying traditional choreographic techniques to corporeal gestures, to engage the relational movement of the body as it intra-acts with a plant-data-body network. Returning to Barad's agential realism, we find intra-actions in the movements of data (sonic and voxel), as it auto-generates through plant signals and passes across bodies (human and plant). Through live performances that improvise with plants as co-composers, we train ourselves to intuitively pay attention to the interferences that matter generates as it passes across bodies or objects. Unique to this work, is Rewa's interpellation of movements from kapa haka, a modality of traditional Māori choreography. Specifically, the wiriwiri hand gesture that imitates the patterns of waves, as Mercier has shown, generates only wavelengths of different frequency [Mercier, 2009]. Wavelengths are visibly diffractive in the medium of water, however through the wiriwiri gesture, are transduced through embodied motion to air. Elucidating connections between wave dynamics in physics and Māori knowledge, is a thread notably explored in scholarship by Williams and Henare [Williams and Hēnare, 2009]. However, wave dynamics are also visibly embedded in other traditional arts such as kōwhaiwhai (painted curvilinear patterns), manifest painted on waka (canoe) paddles, as well as within the panels of whare whakairo (carved meeting houses). Culturally, kōwhaiwhai perform a metaphorical function, since they symbolically transduce precise relational aspects of cosmology that specifically connect to land and bloodline. Framing the eco-digital in *Contact/Sense*, engages a performative interfacing of wave dynamics through wiriwiri, where signals from living plants are translated into data space, using the human body of the performer as a co-creator within a plant led ecology of reciprocal signals.

Through *Contact/Sense*, we highlighted that plants are intra-active through bio-electrical signals, engaging with the affects of light, nutrients and water, and responsive to the physical presence of human bodies, a phenomena also described by Masaoka [Masaoka, 2018]. Plant processes are subatomic and microscopic, and therefore hard to apprehend without sensing apparatus. Broadly speaking, our technique involves sonifying the bio-electrical signals of plants and visualizing that signal as data for the participant in an installation to intra-act with [Figure 2]. More specifically, within each installation, networks that iteratively assemble plants, bodies, bespoke software and hardware devices, afford the capacity for signaletic, computational, and environmental phenomena to emerge. Generated using custom software developed in Unity 3D and Touch Designer, we examined potentials for performative interactions between humans and living plants. The Leap Motion gestural controller was deployed as a camera, a head mounted display (HMD) wearing performer had to look through, affording a de-limited mode of vision that eliminated all color and increased the capacity of the human body to use tactility rather than sight. Tracing the electro-chemical processes of plants through the movement of signal, generates impulsive scales of erratic sounds and rhythms [Figure 3], which Simon Howden improvises alongside in real-time. This installation and performance located human performers in a co-compositional role alongside living plants, whose sonified bio-electrical signal triggered performer improvisations. Crucially, plant signals came first, leading the co-composition and luring humans to follow.



Figure 2: Headset view: Touching one another's signals, human and plant deliver a responsive refrain.

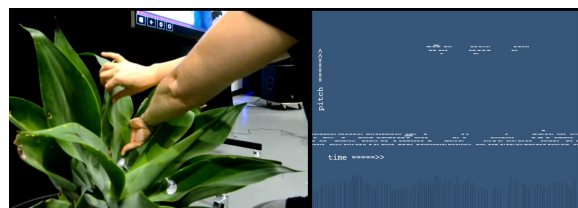


Figure 3: Screen view: LHS; plant-human co-composition. RHS; pitch over time diagram from 'piano roll' window in Logic X.

Reassembling data and human gesture using plant signals contributes to an emerging apprehension that a plant might also be considered as a 'body' in momentum, just operating at a much slower phenological time scale than the human. As a conceptual approach, acknowledging the embodied processes of two modes of situated agency, plant and human, elucidates an understanding of plants as co-creators.

4 NGA MANAWATAKI O TE KOIORA: BIORHYTHMS

Nga manawataki o te koiora: Biorhythms (2022-3, various iterations), is a projection mapping and installation piece that examines the pulse of life, through several intersectional techniques [Wright, 2019]. Firstly, plant signals, transposed to MIDI and blended by Simon Howden with composed music. Secondly, data from those signals, visualized by Rewa Wright using Touch Designer and code in Python. Performed live and then recorded, this work folds together bio-electrical plant sensing with generative data. Conceptually, Rewa developed the curvilinear forms in this piece to evoke her cultural heritage, through a diffractive lens that wove cultural concepts together with computational abstraction. *Nga manawataki o te koiora* leverages poetic motion and audio-reactive data to engulf the viewer within a non-literal computational transduction of the forest, rivers and oceans of Aotearoa/New Zealand [Figure 4]. Displayed in several 320 degree cylinders since 2022, this piece is designed to surround the viewer, who must trace an arc with their body, and turn their head to see the whole work at once [Kohn, 2013]. Visually, an interconnected ecology is translated into the real-time world of audio reactive geometries and mesh topologies. Traditional kete (Māori woven baskets) inspire fluid movements which become pixel topologies. Animated motion made with noise

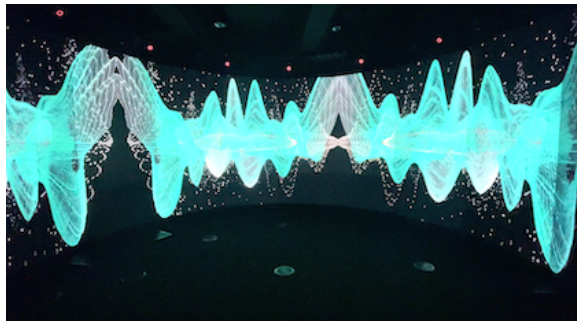


Figure 4: Installation view: *Nga Manawataki o te Koiora* at the EPIcentre, UNSW Sydney, November 2022.

oscillators, shift from reimagined hīnaki (a woven trap) used to catch eel, to seed pods exploding from pixel plants, such as the pōhutukawa (native tree with a red flower). The soft blue/green of kina (sea eggs) become fluffy vectors transparently overlaid on a fluid mesh of waves, breaking apart and transforming into soft lines that converge and melt. Following and modifying the tradition of naturalism and curvilinear geometry that distinguishes Māori artistic expression, this piece visually encapsulates the feeling of the natural world without being a literal representation. Inso doing, it challenges Western pictorial traditions of realism, decolonising the digital tools of computational geometry. Re-assembling those tools through the mode of creation described by the artist as tā rorohiko, this complex audio reactive geometry is further combined with the electrical signals of living plants, who co-create the work [Mercier, 2009]. This 20-minute piece, created in Touch Designer, is sonically a composite of human and plant music, alternately interspersed and sonically mixed. From the ‘human’ side, music consists of three original electronic compositions by Simon Howden. From the plant composition side, we intersperse original recordings of plant bio-electrical signals (ie. biorhythms) captured in research since 2019. We consider plants to be co-composers of this work.

Using computer art to translate emotions about place from bush into pixels, this work explores the potential for a synthesis of data, plants, ecology and algorithms. Mixing the biorhythms of plants with human produced electronic music, reveals a co-creative mesh of human and nonhuman kin, an entanglement that is syncretic to the work itself.

5 TOWARD A SPECULATIVE DECOLONIAL COMPUTATIONAL ART

Woven by radical contingency and perpetual emergence, Indigenous modes of thought open a space for meaning to grow beyond the empirical. They are therefore crucial as strategies to destabilize the ongoing impact of colonial systems of thought and action. This research practice asks: What new ways of intra-acting can we develop alongside computational frameworks, that might afford the capacity to not only reach beyond the human? And, how can performative interfacing, as dynamic and process based, also contribute to the wider project of ideological re-assembly and generate new potentials for decolonial art? Rewa’s small attempts to restore

the dynamic pathways embedded within Indigenous thinking, challenge artificial boundaries such as nature/culture, instead giving respect to concepts of interconnection that ultimately suggest that data itself is only partially empirical. This research contributes to new modalities of human computer interaction that attempt to weave Western epistemology and Indigenous thinking, with the hope of opening an equitable pathway toward an Indigenous Futures in decolonial eco-digital arts practice [Wright, 2022, Smith, 1999].

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Dr. Rewa Wright (Ngai Tawake/Te Uri o Hau/ Te Kaimaroke) is Senior Lecturer in Film, Screen and Animation at the Queensland University of Technology, Australia. Simon Howden is a sound designer and independent music producer. Both humbly reside in Meanjin/Brisbane on the lands of the Turrbal and Yugara, First Nations owners. We recognise that these lands have always been places of teaching, research and learning, and that sovereignty was never ceded.

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